cBioPortal
FOR CANCER GENOMICS

Tutorial 6
Group Comparison

Compare clinical and genomic features of groups of samples

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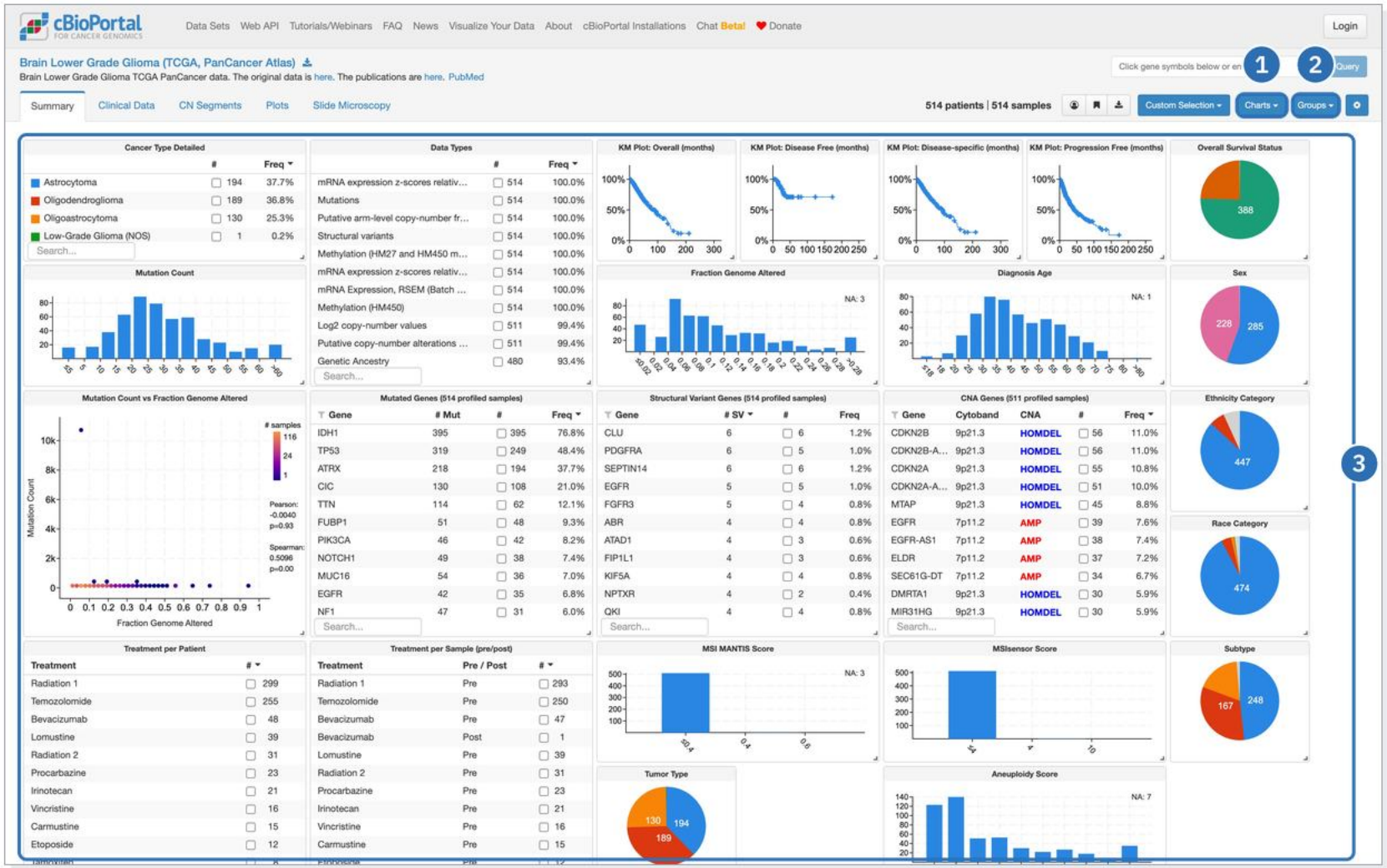
Tutorial objectives

- Explain what the Group Comparison page does
- Show two ways to start a comparison: from a chart in the Study View, or from groups you define
- Walk through the tabs of the Group Comparison page: Overlap, Survival, Clinical, Genomic Alterations and more

What is the Group Comparison page?

- The Group Comparison page compares two or more groups of samples or patients: their survival, clinical attributes, genomic alterations, expression and more, with a statistical test for each.
- Groups come from any chart in the Study View (one group per value, or per range of a number), or from selections you save with the “Groups” button.
- On the page you can also make new groups from the overlap of others.
- The Comparison/Survival tab of the Results View (Tutorial 2) works the same way, with groups made from the query genes.

Starting from a chart: the Study View



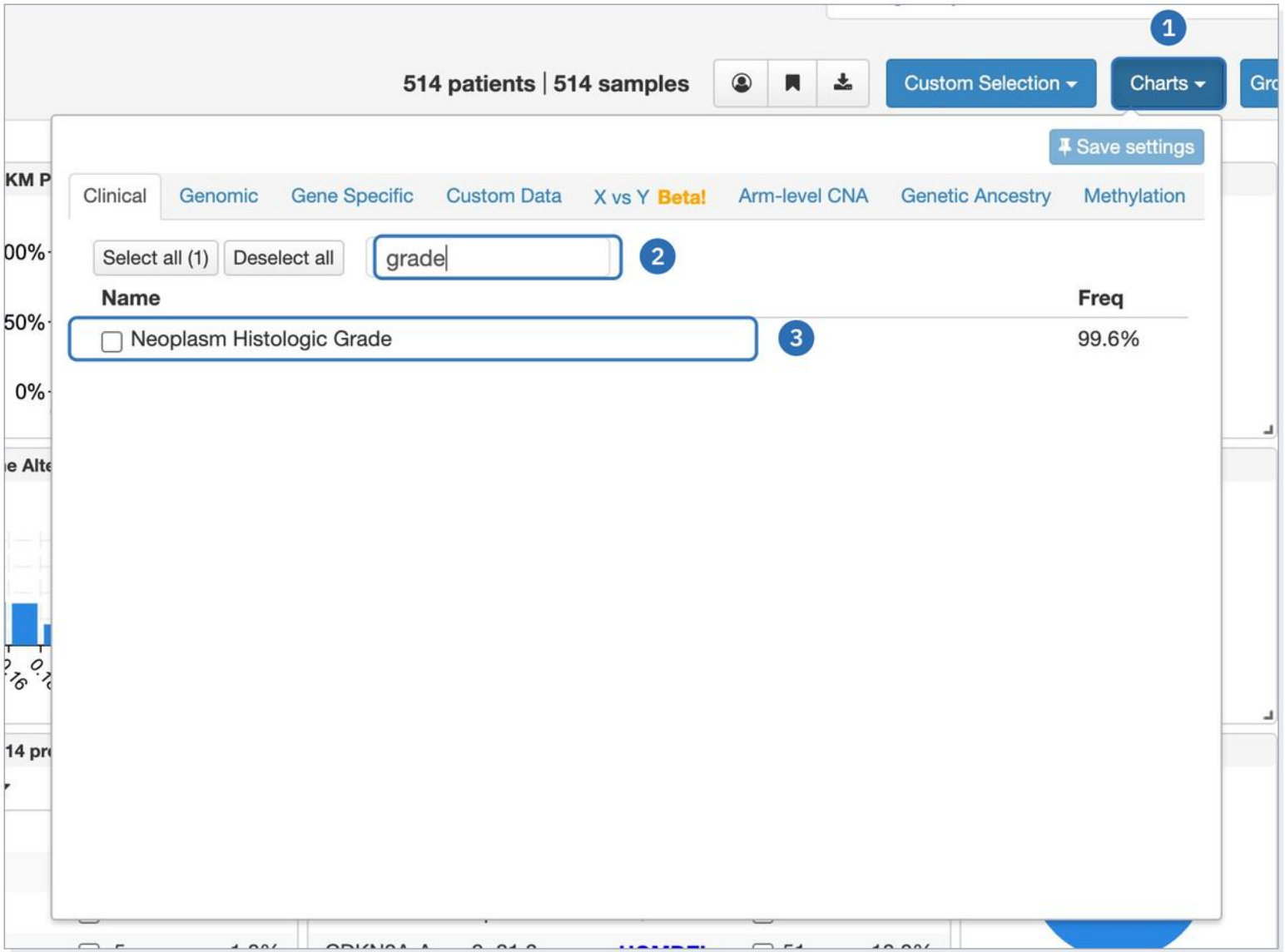
1. The “Charts” menu adds charts. Here we add Neoplasm Histologic Grade, which is not shown by default.

2. The “Groups” button saves selections as groups (second part of this tutorial).

3. Every chart's menu can compare its values as groups.

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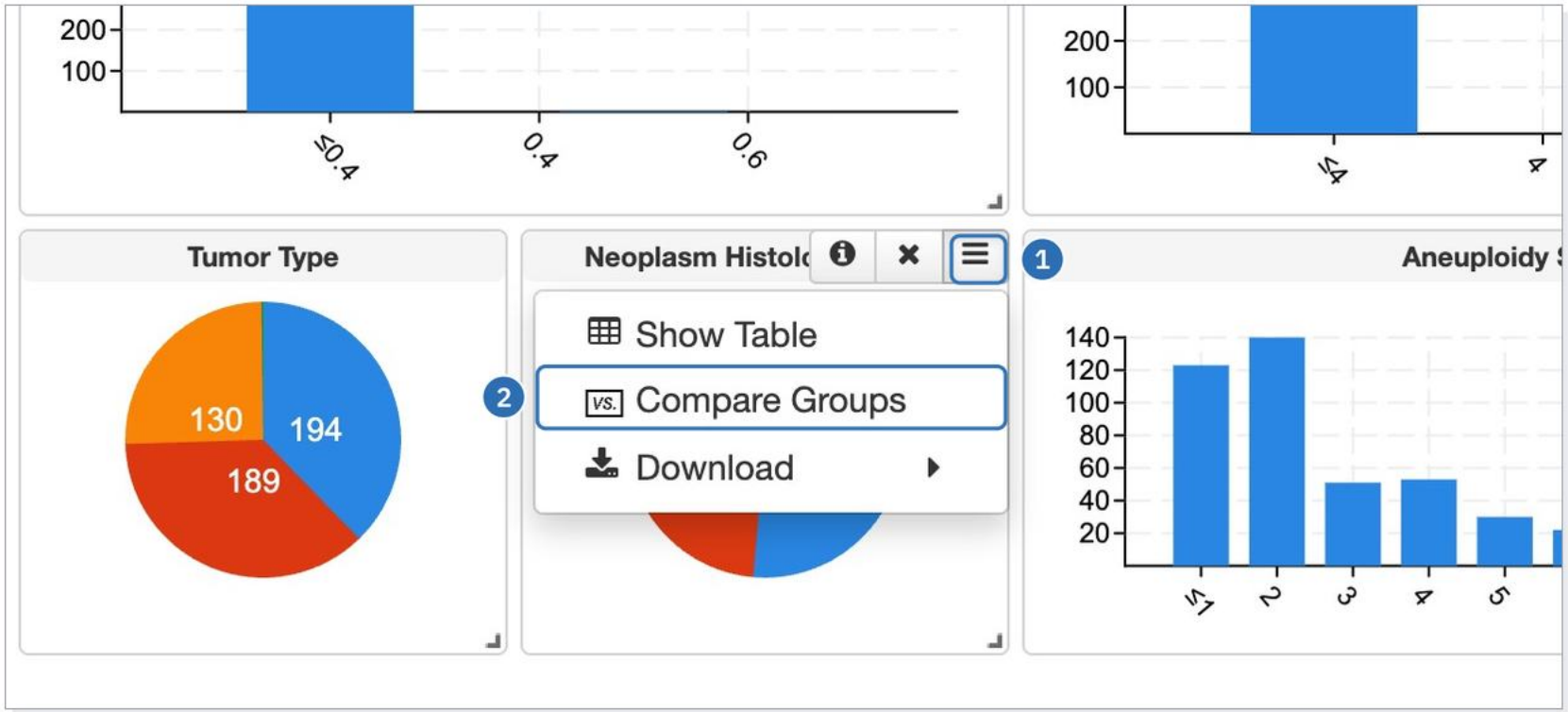
Starting from a chart: add a chart



1. Open the “Charts” menu, Clinical tab.
2. Search for a keyword, e.g. “grade”.
3. Tick “Neoplasm Histologic Grade”. A pie chart is added at the bottom of the Study View.

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Starting from a chart: “Compare Groups”



1. Hover over the chart's menu icon.
2. Click “Compare Groups”: the Group Comparison page opens in a new tab, with one group per value (G2 and G3).

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Group Comparison page: overview



1. The study, and the attribute the groups come from. Click the study name to go back to the Study View.
2. The groups being compared, one per grade, with their sizes. Click a group to deselect it (leave it out of the analysis), its x to remove it; drag to reorder.
3. The tabs. Which ones you see depends on the data in the study.

Group Comparison page: Overlap tab



1. Which samples and patients are in more than one group. G2 and G3 do not overlap.

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Group Comparison page: Survival tab

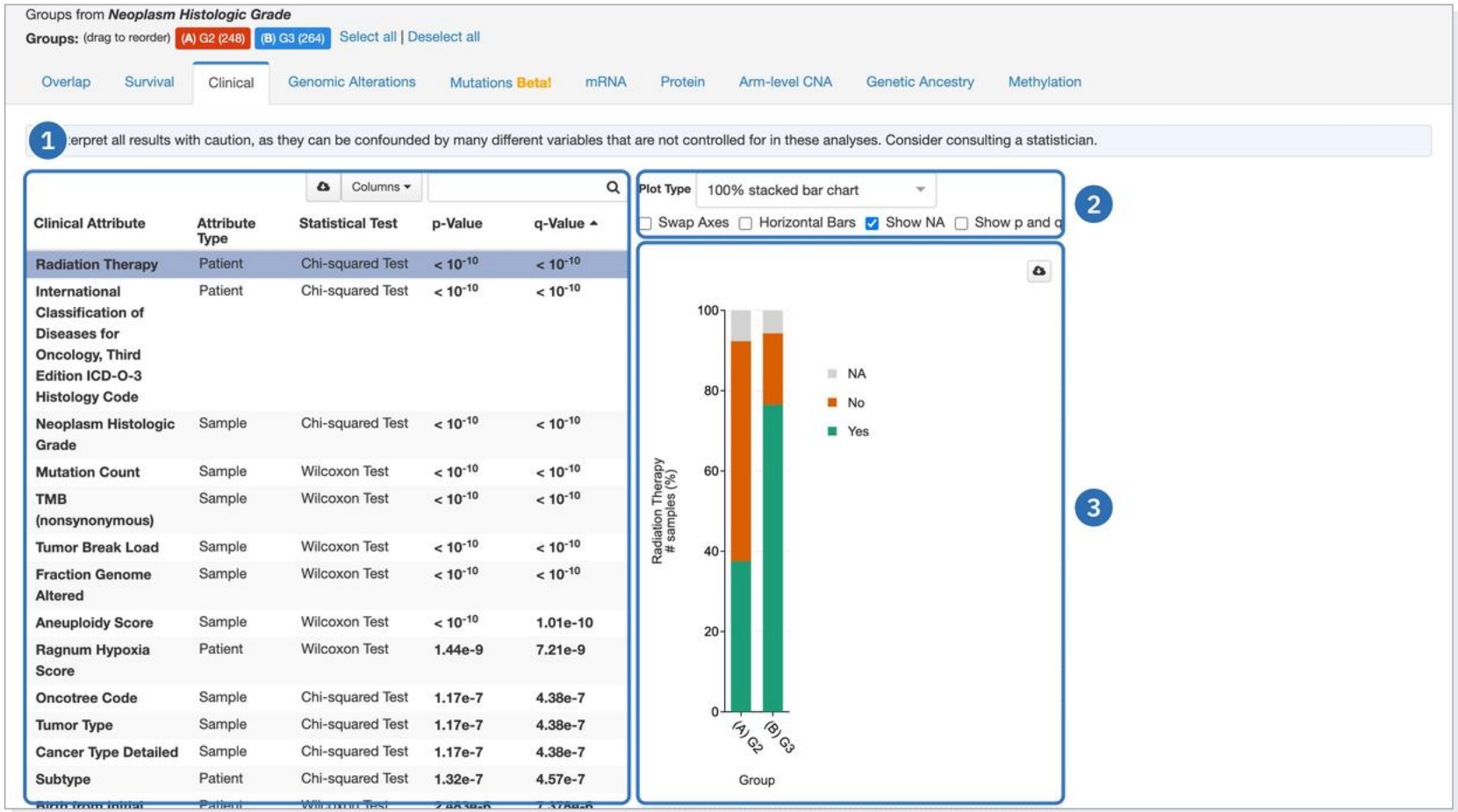


1. The outcomes in the study, with log-rank p- and q-values. Click one to plot it.

2. Q: Do patients with grade 3 tumors have shorter survival?
   A: Yes: the Kaplan-Meier curves show shorter overall survival in G3 (p = 1.65e-10).

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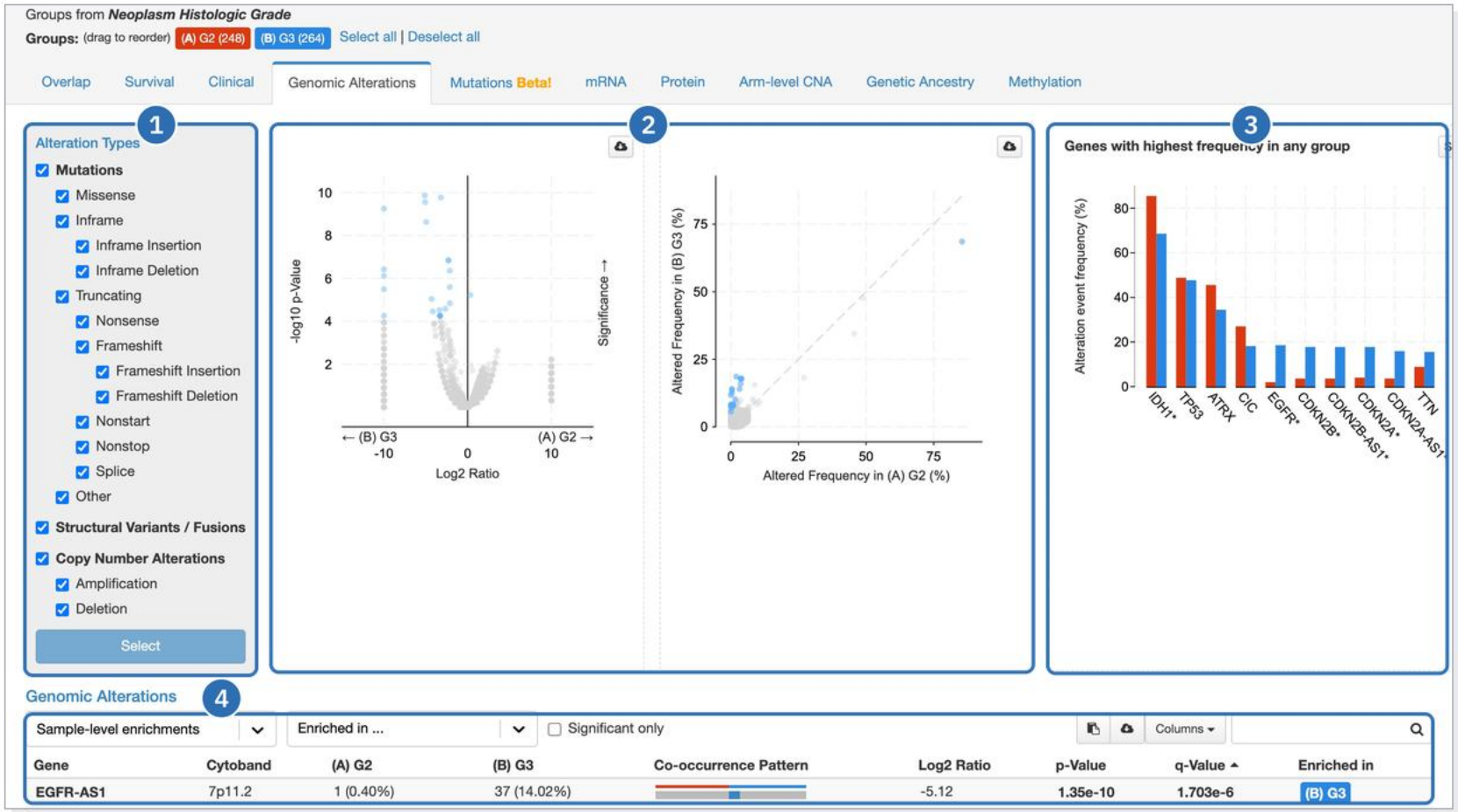
Group Comparison page: Clinical tab



1. Every clinical attribute, tested between the groups, most significant first. Click one to plot it.
2. Options to customize the plot.
3. Values for the selected attribute in each group (here Radiation Therapy).

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Group Comparison page: Genomic Alterations tab



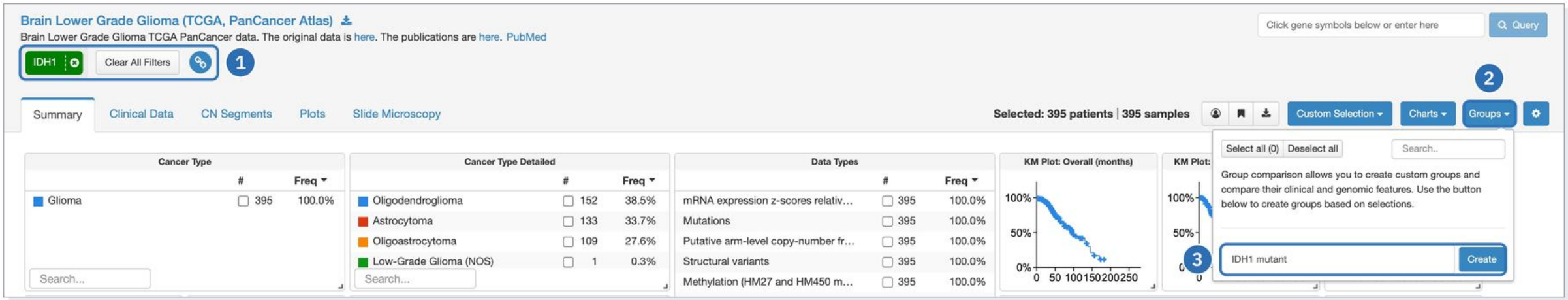
1. Choose which mutations, structural variants and copy-number changes count, then click “Select” to rerun.
2. Overview plots: which genes differ most between the groups, and how often each is altered in each group.
3. The most frequently altered genes in each group.
4. Every gene, tested for enrichment in one group. The mRNA, Protein, Arm-level CNA, Genetic Ancestry and Methylation tabs compare those data in the same way.

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Starting from your own groups

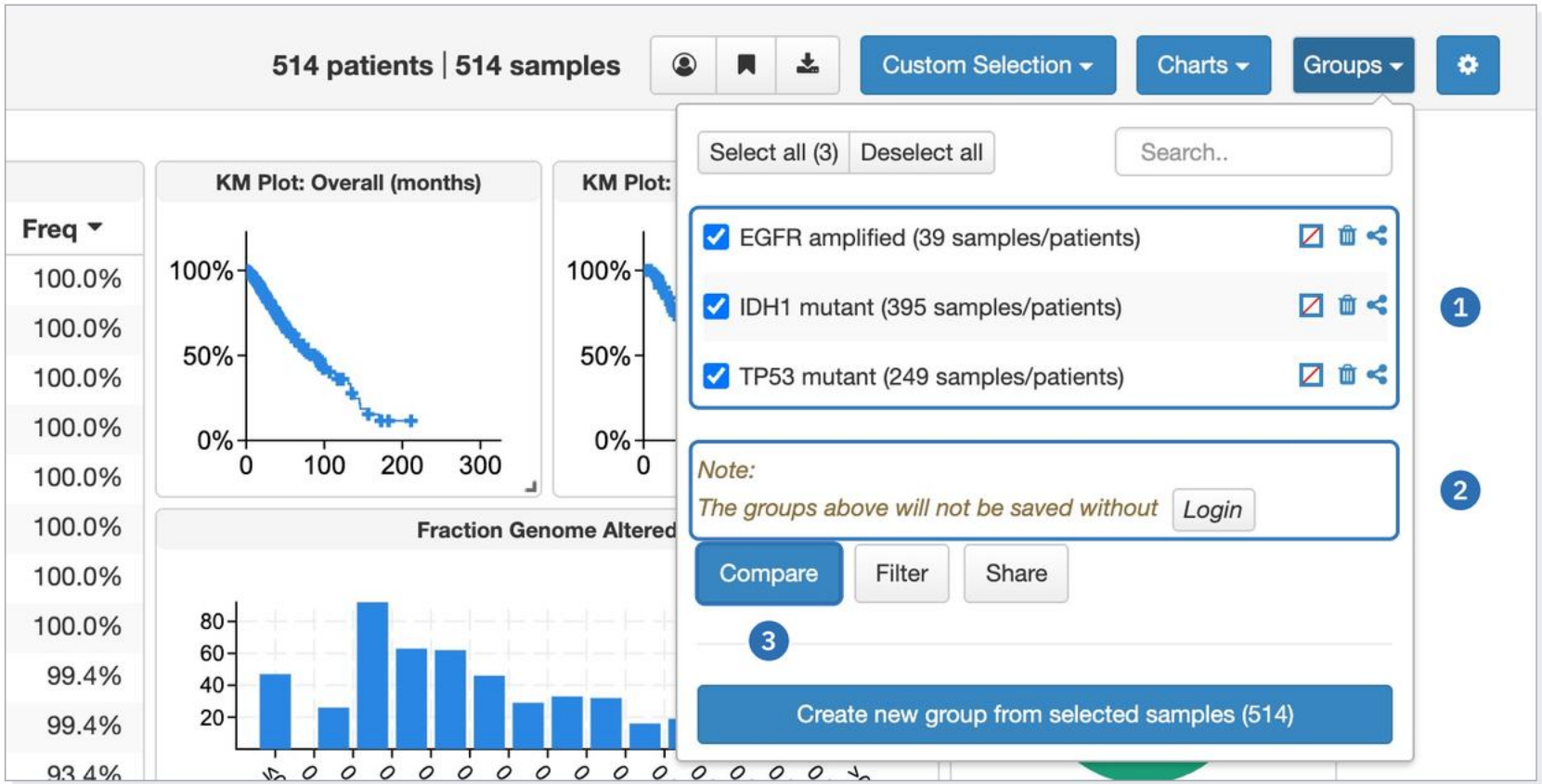
- Any selection in the Study View can be saved as a group with the “Groups” button: filter, then create a group.
- Here: IDH1-mutant samples, TP53-mutant samples and EGFR-amplified samples.
- You don't need to log in to create and compare groups, but without logging in they are lost when the page reloads. Log in to keep them.

Creating a group



1. Filter the Study View: here, samples with an IDH1 mutation (tick IDH1 in the Mutated Genes chart, then “Select Samples”).
2. Open the “Groups” menu and click “Create new group from selected samples”.
3. Name the group (here “IDH1 mutant”) and click “Create”. Then click “Clear All Filters”, filter again and create the next group: TP53 mutant, EGFR amplified (EGFR AMP in the CNA Genes chart).

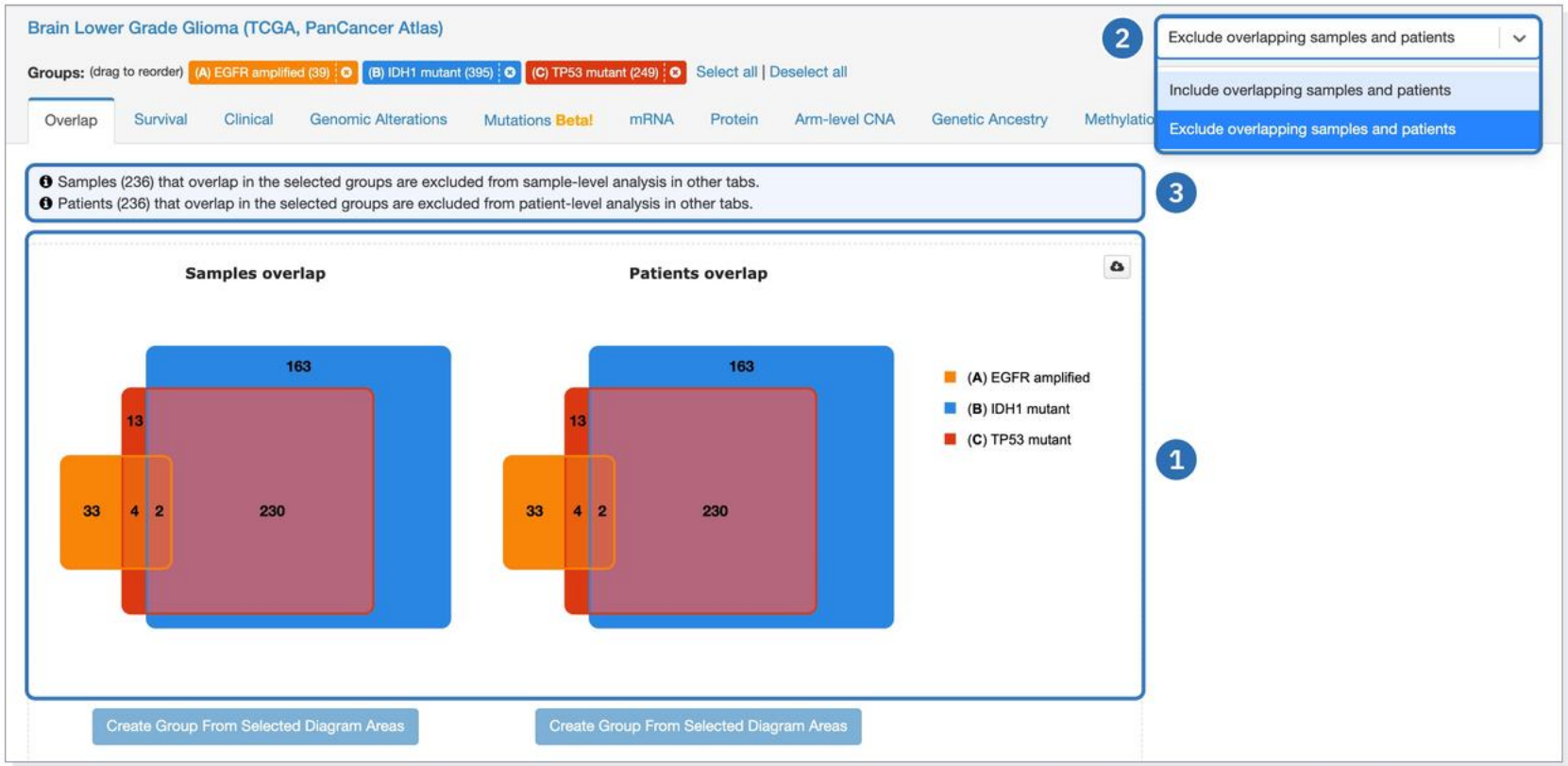
Comparing your groups



1. The groups you saved, with their sizes. Tick them all to compare.
2. Without logging in, groups are kept until the page reloads.
3. Click “Compare”.

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Overlapping groups



1. Unlike the previous example of grades, these groups overlap: the Venn diagram shows that 230 samples are in IDH1 mutant and TP53 mutant only.

2. Choose “Exclude overlapping samples and patients” (the default) or “Include overlapping samples and patients” for the other tabs.

3. Each tab says how many overlapping samples and patients are left out (here 236).

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A new group from the overlap



1. Click an area of the diagram to select it: here, samples with IDH1 and TP53 mutations and no EGFR amplification (230). Click more areas to add them.
2. Name the group ("IDH1 + TP53, no EGFR amp") and click "Submit".
3. "Create Group From Selected Diagram Areas" opens the name box. The comparison reloads with the new group. When logged in, there is also an option to save the group to your profile: it is then available in the Study View for this study.

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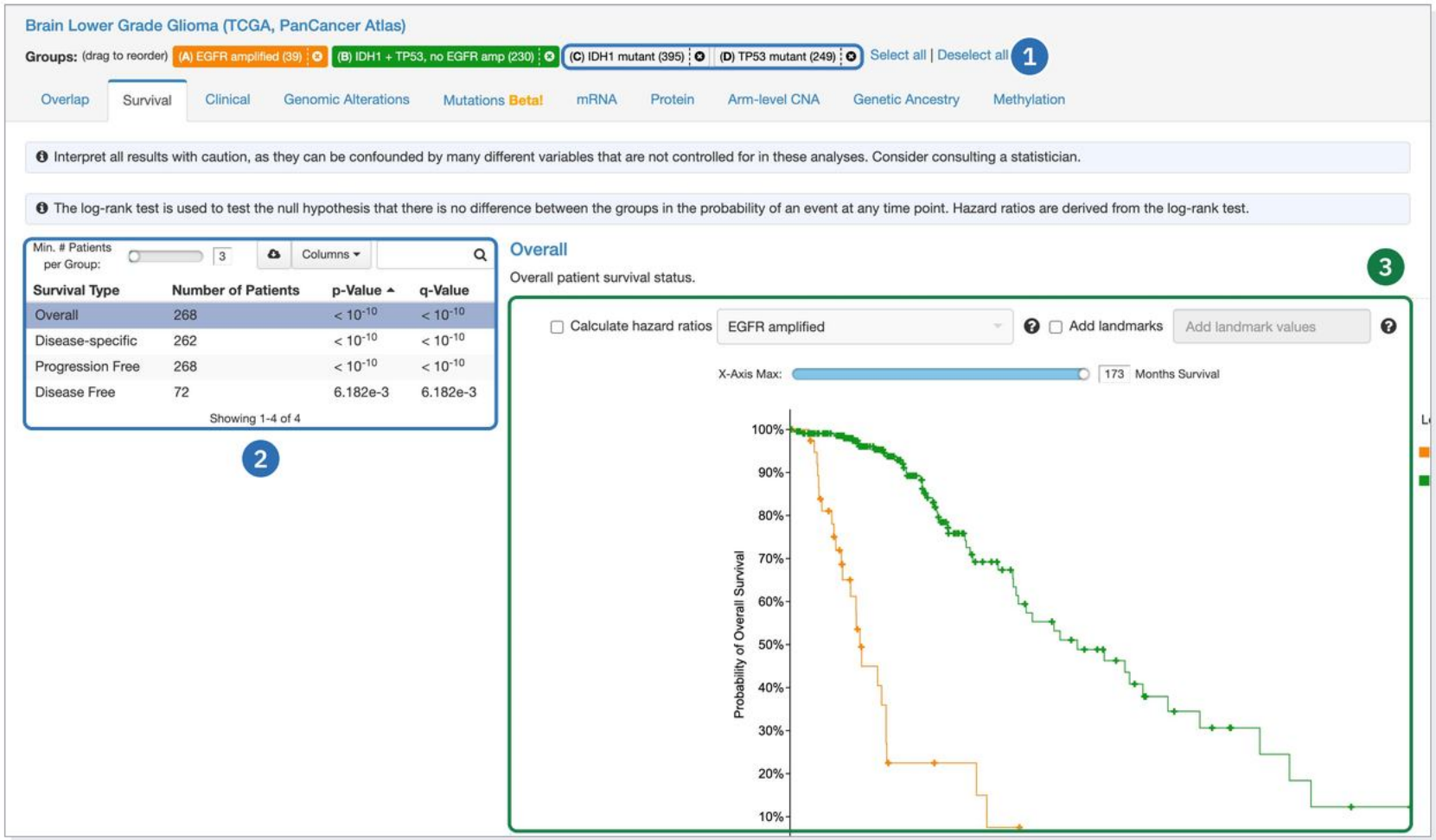
Four groups or more: the UpSet plot



1. The new group (B) is added.
2. It lies entirely inside other groups, so it is left out of the other tabs until you deselect them.
3. With four or more groups the overlap is an UpSet plot: each bar is one combination of groups, shown by the dots below.

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Comparing outcomes of your groups



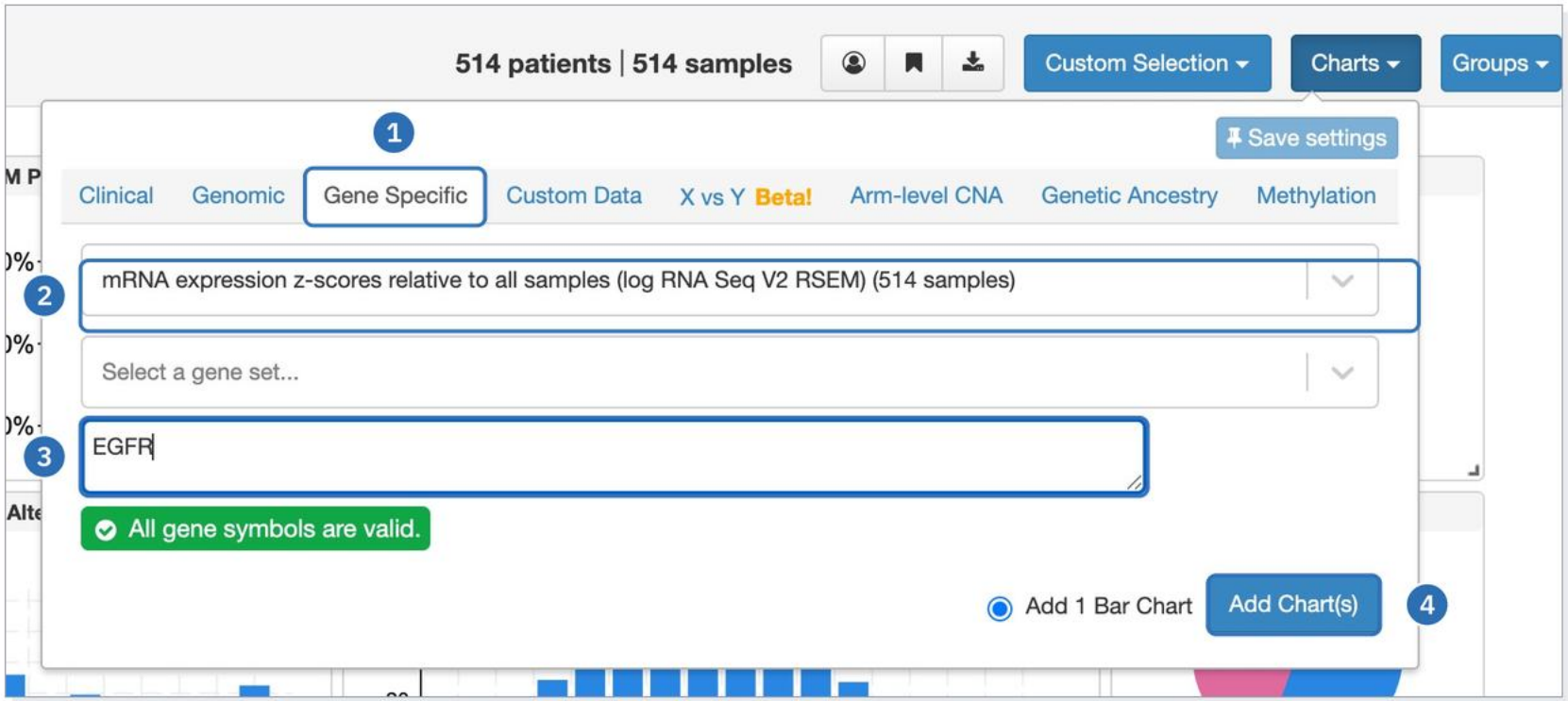
1. Click “IDH1 mutant” and “TP53 mutant” to deselect them.
2. Overall, disease-specific and progression-free survival all differ ($p < 1e\text{-}10$).
3. Q: Do patients with EGFR amplification have a different outcome from those with IDH1 and TP53 mutations?
   A: Yes: overall survival is much shorter with EGFR amplification.

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Example: high vs low EGFR expression

- EGFR is often gained or amplified in gliomas, and then expressed at high levels.
- Q: Do patients whose tumors express EGFR highly have a different outcome from those with low EGFR expression?
- Add a gene-specific chart of EGFR mRNA, split it into quartiles, and compare the top quartile with the bottom one.

EGFR expression: add a gene-specific chart



1. "Charts" menu, Gene Specific tab.
2. Choose which expression data to plot: here z-scores, which show how high or low each sample's EGFR expression is compared with all samples.
3. Enter your gene of interest, here EGFR.
4. Click "Add Chart(s)": a histogram of EGFR expression is added to the Study View.

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EGFR expression: two ways to make groups



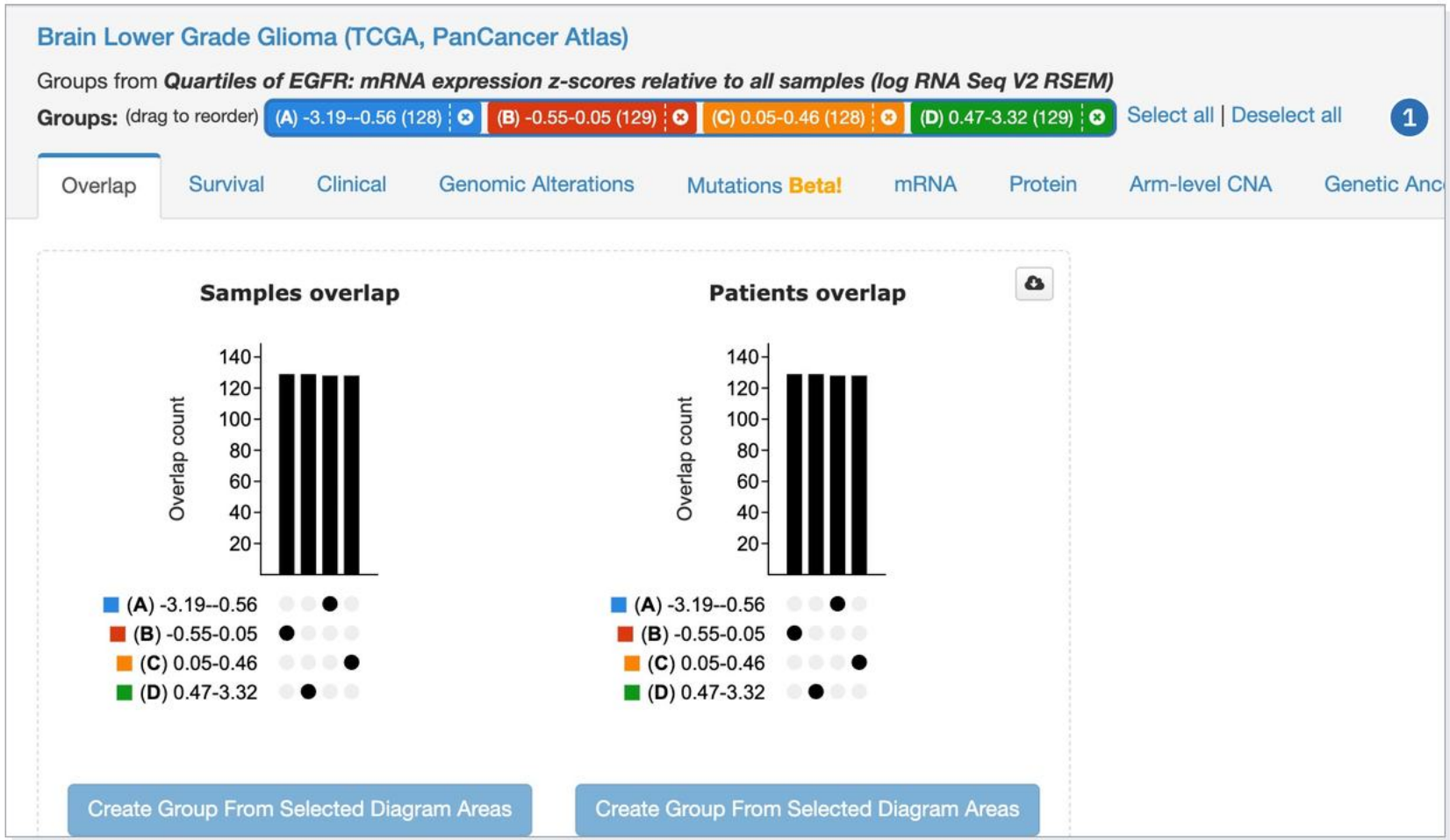
1. Option 1: click or drag on the bars to select a range of EGFR expression, save it as a group with the “Groups” button, do the same for another range, then compare the groups.

2. Option 2, used here: hover over the chart's menu icon and choose “Compare Groups”...

3. ...then how to split the samples. Here we pick “Quartiles” (four groups of about 128 samples). “Median” makes two groups, “Current bins” one group per bar, and “Custom Bins” sets your own bin boundaries first.

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EGFR expression: the quartile groups



1 Four groups, from (A) the lowest EGFR expression (z-score -3.19 to -0.56) to (D) the highest (0.47 to 3.32). They do not overlap. Click (B) and (C) to deselect them: the next tabs compare the top and bottom quartiles.

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EGFR expression: Survival tab

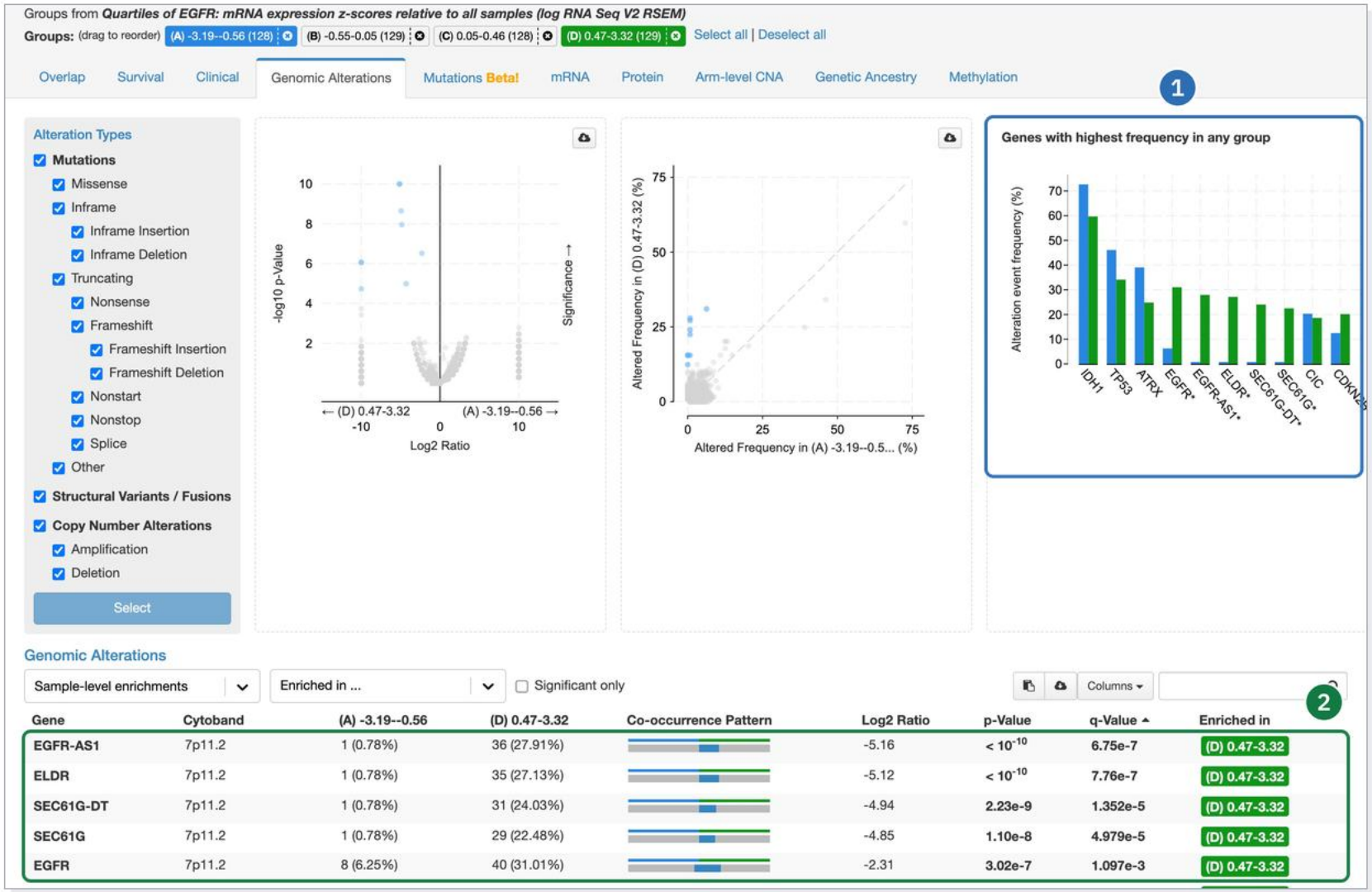


1. (B) and (C) are deselected (white).

2. Each outcome is tested between the groups. Disease-specific and progression-free survival differ ($p < 0.05$), overall survival doesn't, and after correcting for testing four outcomes none is significant.

3. Q: Do patients with high EGFR expression have a different outcome?
   A: Survival tends to be shorter in the highest quartile (D, green) than in the lowest (A, blue), but the difference is modest and not significant after correction for the four outcomes tested.

EGFR expression: Genomic Alterations tab



1. Genes most often altered in either group: IDH1, TP53 and ATRX are more frequent in the low group; EGFR and its neighbors on 7p11.2 in the high group.

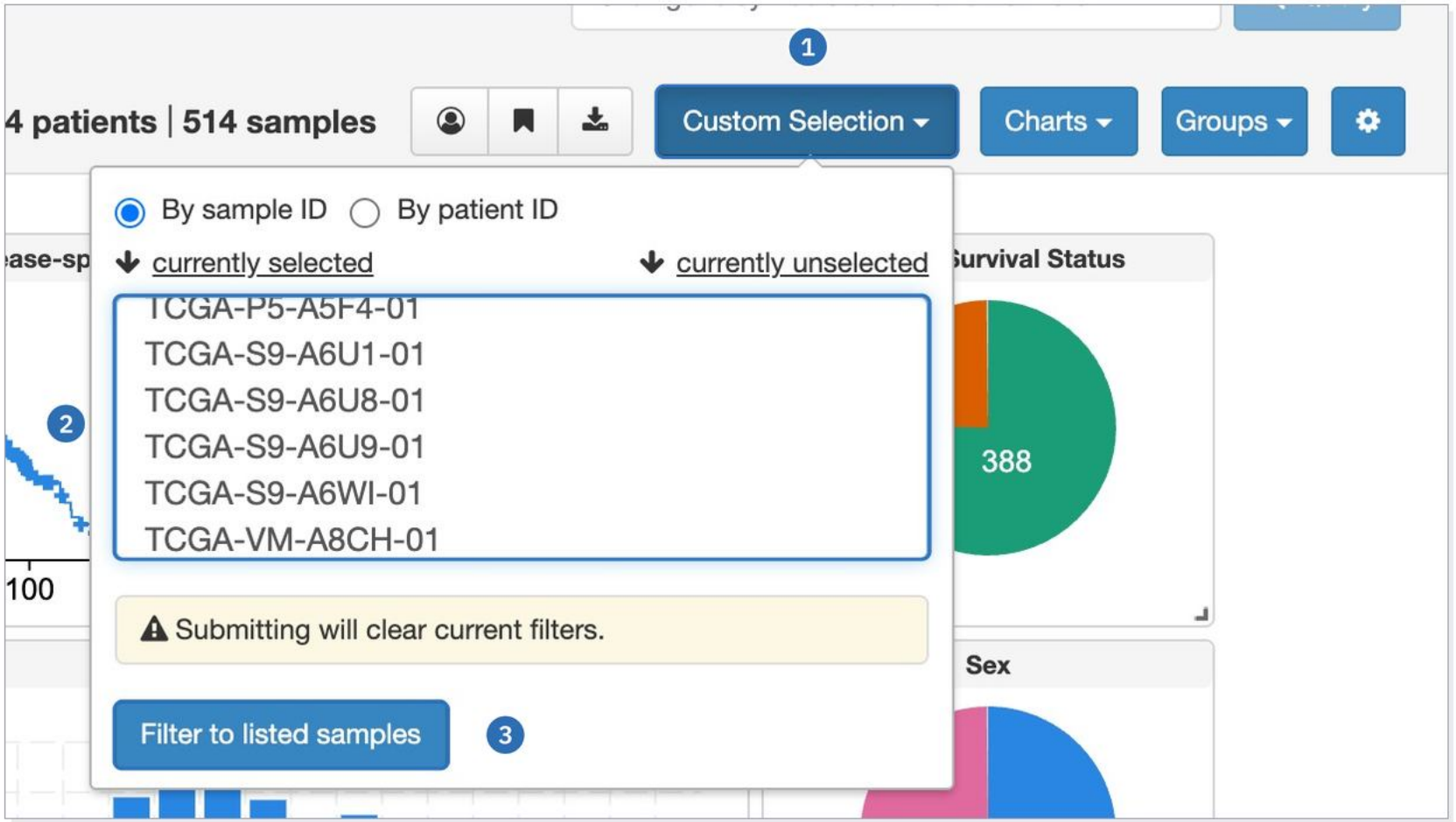
2. Q: Why do some tumors express EGFR highly?
A: EGFR is altered in 31% of the high group vs 6% of the low group (q = 0.001), and so are the genes next to it on 7p11.2 (EGFR-AS1, ELDR, SEC61G). Alterations shared with neighboring genes point to amplification: high expression often comes from extra copies of EGFR.

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Another way to make a group: your own list of samples

- Groups don't have to come from charts or filters in the Study View.
- If you already have a list of samples, from your own analysis or from the Download tab of the Results View, paste it to select those samples, then save them as a group and compare it like any other.

A group from your own list of samples



1. Open “Custom Selection”.
2. Paste the sample IDs, one per line, or choose patient IDs. Here, a list of 17 samples.
3. Click “Filter to listed samples”, then create a group from the selection with the “Groups” button.

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Questions?

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